

# CoP SAP 6.2 Appendix

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## **6.2 Appendix Correlations of Virus-Specific Antibodies and Viral Load by Sex**

We recalculate some of the Tables from CoP SAP 6.2 by sex.

## Virus-Specific Antibodies

### Correlations

Simple statistics will be performed independent of the vaccine product. For example, the Spearman correlation between each pair of virus-specific antibodies will be calculated. Spearman (i.e., rank) correlations with confidence intervals will be generated; confidence intervals for the Spearman's correlation coefficient are calculated by using the ranks in a Pearson's correlation, which generally performs well (Caruso and Cliff, 1997). As there were not many ties, nonparametric bootstrap as a sensitivity analysis was not done.

Table 6.2.1m: (Male Only) Spearman Correlations of Virus-Specific Antibodies

|                                                 | log<br>MN 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>MN<br>day<br>chal-<br>lenge | log<br>MSD<br>ECL 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>MSD<br>ECL<br>day<br>chal-<br>lenge | log<br>PsVNA<br>ID50 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>PsVNA<br>ID50<br>day<br>chal-<br>lenge | log<br>PsVNA<br>ID80 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>PsVNA<br>ID80<br>day<br>chal-<br>lenge |
|-------------------------------------------------|-------------------------------------------------|------------------------------------|---------------------------------------------------------|--------------------------------------------|------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|-----------------------------------------------|
| log MN 2<br>weeks prior<br>challenge            | -                                               | -                                  | -                                                       | -                                          | -                                                          | -                                             | -                                                          | -                                             |
| log MN day<br>challenge                         | 0.89<br>(0.82,<br>0.93)                         | -                                  | -                                                       | -                                          | -                                                          | -                                             | -                                                          | -                                             |
| log MSD<br>ECL 2<br>weeks prior<br>challenge    | 0.92<br>(0.87,<br>0.95)                         | 0.89<br>(0.83,<br>0.94)            | -                                                       | -                                          | -                                                          | -                                             | -                                                          | -                                             |
| log MSD<br>ECL day<br>challenge                 | 0.91<br>(0.86,<br>0.95)                         | 0.9<br>(0.84,<br>0.94)             | 0.98<br>(0.97,<br>0.99)                                 | -                                          | -                                                          | -                                             | -                                                          | -                                             |
| log PsVNA<br>ID50 2<br>weeks prior<br>challenge | 0.91<br>(0.86,<br>0.95)                         | 0.87<br>(0.79,<br>0.92)            | 0.92<br>(0.87,<br>0.95)                                 | 0.91<br>(0.85,<br>0.94)                    | -                                                          | -                                             | -                                                          | -                                             |
| log PsVNA<br>ID50 day<br>challenge              | 0.89<br>(0.82,<br>0.93)                         | 0.84<br>(0.76,<br>0.9)             | 0.91<br>(0.85,<br>0.94)                                 | 0.9<br>(0.83,<br>0.94)                     | 0.94<br>(0.9,<br>0.96)                                     | -                                             | -                                                          | -                                             |
| log PsVNA<br>ID80 2<br>weeks prior<br>challenge | 0.93<br>(0.89,<br>0.96)                         | 0.87<br>(0.79,<br>0.92)            | 0.93<br>(0.89,<br>0.96)                                 | 0.91<br>(0.86,<br>0.95)                    | 0.97<br>(0.95,<br>0.98)                                    | 0.92<br>(0.87,<br>0.95)                       | -                                                          | -                                             |
| log PsVNA<br>ID80 day<br>challenge              | 0.93<br>(0.88,<br>0.96)                         | 0.9<br>(0.84,<br>0.94)             | 0.94<br>(0.9,<br>0.96)                                  | 0.94<br>(0.91,<br>0.97)                    | 0.95<br>(0.91,<br>0.97)                                    | 0.94<br>(0.9,<br>0.96)                        | 0.95<br>(0.92,<br>0.97)                                    | -                                             |

6.2.1f: (Female Only) Spearman Correlations of Virus-Specific Antibodies

|                                                 | log<br>MN 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>MN<br>day<br>chal-<br>lenge | log<br>MSD<br>ECL 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>MSD<br>ECL<br>day<br>chal-<br>lenge | log<br>PsVNA<br>ID50 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>PsVNA<br>ID50<br>day<br>chal-<br>lenge | log<br>PsVNA<br>ID80 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>PsVNA<br>ID80<br>day<br>chal-<br>lenge |
|-------------------------------------------------|-------------------------------------------------|------------------------------------|---------------------------------------------------------|--------------------------------------------|------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|-----------------------------------------------|
| log MN 2<br>weeks prior<br>challenge            | -                                               | -                                  | -                                                       | -                                          | -                                                          | -                                             | -                                                          | -                                             |
| log MN day<br>challenge                         | 0.91<br>(0.85,<br>0.94)                         | -                                  | -                                                       | -                                          | -                                                          | -                                             | -                                                          | -                                             |
| log MSD<br>ECL 2<br>weeks prior<br>challenge    | 0.93<br>(0.89,<br>0.96)                         | 0.88<br>(0.81,<br>0.93)            | -                                                       | -                                          | -                                                          | -                                             | -                                                          | -                                             |
| log MSD<br>ECL day<br>challenge                 | 0.92<br>(0.88,<br>0.95)                         | 0.88<br>(0.81,<br>0.93)            | 0.97<br>(0.95,<br>0.98)                                 | -                                          | -                                                          | -                                             | -                                                          | -                                             |
| log PsVNA<br>ID50 2<br>weeks prior<br>challenge | 0.88<br>(0.81,<br>0.93)                         | 0.85<br>(0.77,<br>0.91)            | 0.9<br>(0.84,<br>0.94)                                  | 0.87<br>(0.79,<br>0.92)                    | -                                                          | -                                             | -                                                          | -                                             |
| log PsVNA<br>ID50 day<br>challenge              | 0.87<br>(0.79,<br>0.92)                         | 0.87<br>(0.8,<br>0.92)             | 0.89<br>(0.82,<br>0.93)                                 | 0.88<br>(0.81,<br>0.93)                    | 0.92<br>(0.87,<br>0.95)                                    | -                                             | -                                                          | -                                             |
| log PsVNA<br>ID80 2<br>weeks prior<br>challenge | 0.91<br>(0.86,<br>0.95)                         | 0.88<br>(0.82,<br>0.93)            | 0.93<br>(0.89,<br>0.96)                                 | 0.92<br>(0.86,<br>0.95)                    | 0.95<br>(0.92,<br>0.97)                                    | 0.92<br>(0.87,<br>0.95)                       | -                                                          | -                                             |
| log PsVNA<br>ID80 day<br>challenge              | 0.92<br>(0.88,<br>0.95)                         | 0.93<br>(0.88,<br>0.95)            | 0.92<br>(0.87,<br>0.95)                                 | 0.93<br>(0.89,<br>0.96)                    | 0.92<br>(0.86,<br>0.95)                                    | 0.94<br>(0.91,<br>0.96)                       | 0.94<br>(0.9,<br>0.96)                                     | -                                             |

## Viral Load

### Correlations - Numeric Viral Load

Spearman's correlations of all numeric viral load measurements with each other will be calculated.

6.2.2m: (Male Only) Spearman Correlations of Viral Load Outcomes

|                                                           | AUC<br>BAL<br>N1        | AUC<br>BAL<br>Subge-<br>nomic | AUC<br>Nasal<br>Swab<br>N1 | AUC<br>Nasal<br>Swab<br>Subge-<br>nomic | AUC<br>OP<br>Swab<br>N1 | AUC<br>OP<br>Swab<br>Subge-<br>nomic | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>BAL<br>N1 | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>BAL<br>Subge-<br>nomic | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>OP-<br>swab<br>N1 | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>OP-<br>swab<br>Subge-<br>nomic | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>Nasal-<br>Swab<br>N1 | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>Nasal-<br>Swab<br>Subge-<br>nomic |
|-----------------------------------------------------------|-------------------------|-------------------------------|----------------------------|-----------------------------------------|-------------------------|--------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------|
| AUC BAL<br>N1                                             | -                       | -                             | -                          | -                                       | -                       | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| AUC BAL<br>Subgenomic                                     | 0.91<br>(0.86,<br>0.95) | -                             | -                          | -                                       | -                       | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| AUC Nasal<br>Swab N1                                      | 0.63<br>(0.45,<br>0.76) | 0.61<br>(0.43,<br>0.75)       | -                          | -                                       | -                       | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| AUC Nasal<br>Swab<br>Subgenomic                           | 0.6<br>(0.42,<br>0.74)  | 0.57<br>(0.37,<br>0.72)       | 0.83<br>(0.74,<br>0.89)    | -                                       | -                       | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| AUC OP<br>Swab N1                                         | 0.8<br>(0.69,<br>0.88)  | 0.75<br>(0.62,<br>0.84)       | 0.8<br>(0.69,<br>0.87)     | 0.71<br>(0.56,<br>0.81)                 | -                       | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| AUC OP<br>Swab<br>Subgenomic                              | 0.74<br>(0.6,<br>0.83)  | 0.69<br>(0.53,<br>0.8)        | 0.69<br>(0.54,<br>0.8)     | 0.63<br>(0.46,<br>0.76)                 | 0.89<br>(0.83,<br>0.93) | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| log VL Day2<br>PostChal-<br>lenge BAL<br>N1               | 0.88<br>(0.81,<br>0.93) | 0.91<br>(0.85,<br>0.94)       | 0.54<br>(0.34,<br>0.69)    | 0.49<br>(0.28,<br>0.66)                 | 0.7<br>(0.55,<br>0.81)  | 0.6<br>(0.42,<br>0.74)               | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| log VL Day2<br>PostChal-<br>lenge BAL<br>Subgenomic       | 0.84<br>(0.74,<br>0.9)  | 0.94<br>(0.9,<br>0.96)        | 0.54<br>(0.33,<br>0.69)    | 0.49<br>(0.28,<br>0.66)                 | 0.68<br>(0.52,<br>0.8)  | 0.6<br>(0.41,<br>0.74)               | 0.95<br>(0.91,<br>0.97)                                  | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| log VL Day2<br>PostChal-<br>lenge<br>OPswab N1            | 0.46<br>(0.24,<br>0.63) | 0.4<br>(0.17,<br>0.59)        | 0.38<br>(0.15,<br>0.58)    | 0.43<br>(0.21,<br>0.61)                 | 0.52<br>(0.32,<br>0.68) | 0.55<br>(0.35,<br>0.7)               | 0.35<br>(0.11,<br>0.55)                                  | 0.34<br>(0.1,<br>0.54)                                                | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| log VL Day2<br>PostChal-<br>lenge<br>OPswab<br>Subgenomic | 0.46<br>(0.24,<br>0.63) | 0.42<br>(0.19,<br>0.61)       | 0.42<br>(0.19,<br>0.6)     | 0.48<br>(0.26,<br>0.65)                 | 0.57<br>(0.37,<br>0.72) | 0.61<br>(0.43,<br>0.75)              | 0.34<br>(0.1,<br>0.54)                                   | 0.35<br>(0.12,<br>0.55)                                               | 0.96<br>(0.93,<br>0.97)                                          | -                                                                             | -                                                                   | -                                                                                |

|             |        |        |        |        |        |        |        |        |        |        |        |   |
|-------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---|
| log VL Day2 | 0.57   | 0.56   | 0.69   | 0.77   | 0.73   | 0.6    | 0.54   | 0.55   | 0.48   | 0.52   | -      | - |
| PostChal-   | (0.37, | (0.37, | (0.53, | (0.64, | (0.59, | (0.41, | (0.34, | (0.35, | (0.26, | (0.31, |        |   |
| lenge       | 0.71)  | 0.71)  | 0.8)   | 0.85)  | 0.83)  | 0.74)  | 0.7)   | 0.7)   | 0.65)  | 0.68)  |        |   |
| NasalSwab   |        |        |        |        |        |        |        |        |        |        |        |   |
| N1          |        |        |        |        |        |        |        |        |        |        |        |   |
| log VL Day2 | 0.54   | 0.56   | 0.68   | 0.79   | 0.7    | 0.59   | 0.53   | 0.56   | 0.44   | 0.49   | 0.97   | - |
| PostChal-   | (0.34, | (0.36, | (0.51, | (0.67, | (0.55, | (0.41, | (0.32, | (0.36, | (0.22, | (0.28, | (0.95, |   |
| lenge       | 0.7)   | 0.71)  | 0.79)  | 0.87)  | 0.81)  | 0.73)  | 0.69)  | 0.71)  | 0.62)  | 0.66)  | 0.98)  |   |
| NasalSwab   |        |        |        |        |        |        |        |        |        |        |        |   |
| Subgenomic  |        |        |        |        |        |        |        |        |        |        |        |   |

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6.2.2f: (Female Only) Spearman Correlations of Viral Load Outcomes

|                                                           | AUC<br>BAL<br>N1        | AUC<br>BAL<br>Subge-<br>nomic | AUC<br>Nasal<br>Swab<br>N1 | AUC<br>Nasal<br>Swab<br>Subge-<br>nomic | AUC<br>OP<br>Swab<br>N1 | AUC<br>OP<br>Swab<br>Subge-<br>nomic | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>BAL<br>N1 | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>BAL<br>Subge-<br>nomic | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>OP-<br>swab<br>N1 | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>OP-<br>swab<br>Subge-<br>nomic | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>Nasal-<br>Swab<br>N1 | log<br>VL<br>Day2<br>Post<br>Chal-<br>lenge<br>Nasal-<br>Swab<br>Subge-<br>nomic |
|-----------------------------------------------------------|-------------------------|-------------------------------|----------------------------|-----------------------------------------|-------------------------|--------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------|
| AUC BAL<br>N1                                             | -                       | -                             | -                          | -                                       | -                       | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| AUC BAL<br>Subgenomic                                     | 0.83<br>(0.73,<br>0.89) | -                             | -                          | -                                       | -                       | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| AUC Nasal<br>Swab N1                                      | 0.61<br>(0.42,<br>0.74) | 0.59<br>(0.4,<br>0.73)        | -                          | -                                       | -                       | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| AUC Nasal<br>Swab<br>Subgenomic                           | 0.61<br>(0.43,<br>0.75) | 0.52<br>(0.32,<br>0.68)       | 0.89<br>(0.83,<br>0.93)    | -                                       | -                       | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| AUC OP<br>Swab N1                                         | 0.66<br>(0.49,<br>0.78) | 0.49<br>(0.27,<br>0.66)       | 0.76<br>(0.63,<br>0.85)    | 0.79<br>(0.67,<br>0.87)                 | -                       | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| AUC OP<br>Swab<br>Subgenomic                              | 0.66<br>(0.49,<br>0.78) | 0.52<br>(0.31,<br>0.68)       | 0.71<br>(0.56,<br>0.82)    | 0.74<br>(0.6,<br>0.84)                  | 0.9<br>(0.84,<br>0.94)  | -                                    | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| log VL Day2<br>PostChal-<br>lenge BAL<br>N1               | 0.85<br>(0.76,<br>0.91) | 0.83<br>(0.74,<br>0.9)        | 0.44<br>(0.22,<br>0.62)    | 0.45<br>(0.23,<br>0.63)                 | 0.47<br>(0.25,<br>0.65) | 0.51<br>(0.3,<br>0.68)               | -                                                        | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| log VL Day2<br>PostChal-<br>lenge BAL<br>Subgenomic       | 0.8<br>(0.69,<br>0.88)  | 0.92<br>(0.88,<br>0.95)       | 0.45<br>(0.23,<br>0.63)    | 0.44<br>(0.21,<br>0.62)                 | 0.45<br>(0.22,<br>0.63) | 0.48<br>(0.26,<br>0.65)              | 0.9<br>(0.84,<br>0.94)                                   | -                                                                     | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| log VL Day2<br>PostChal-<br>lenge<br>OPswab N1            | 0.41<br>(0.18,<br>0.6)  | 0.31<br>(0.07,<br>0.52)       | 0.52<br>(0.32,<br>0.68)    | 0.47<br>(0.25,<br>0.64)                 | 0.53<br>(0.33,<br>0.69) | 0.61<br>(0.43,<br>0.75)              | 0.28<br>(0.03,<br>0.5)                                   | 0.19<br>(-<br>0.06,<br>0.42)                                          | -                                                                | -                                                                             | -                                                                   | -                                                                                |
| log VL Day2<br>PostChal-<br>lenge<br>OPswab<br>Subgenomic | 0.46<br>(0.24,<br>0.63) | 0.37<br>(0.14,<br>0.57)       | 0.59<br>(0.41,<br>0.73)    | 0.52<br>(0.31,<br>0.68)                 | 0.55<br>(0.35,<br>0.7)  | 0.66<br>(0.5,<br>0.78)               | 0.33<br>(0.08,<br>0.53)                                  | 0.25<br>(0,<br>0.47)                                                  | 0.97<br>(0.95,<br>0.98)                                          | -                                                                             | -                                                                   | -                                                                                |
| log VL Day2<br>PostChal-<br>lenge<br>NasalSwab<br>N1      | 0.53<br>(0.33,<br>0.69) | 0.39<br>(0.16,<br>0.58)       | 0.74<br>(0.6,<br>0.83)     | 0.75<br>(0.62,<br>0.84)                 | 0.62<br>(0.44,<br>0.75) | 0.61<br>(0.42,<br>0.74)              | 0.38<br>(0.15,<br>0.58)                                  | 0.29<br>(0.05,<br>0.51)                                               | 0.49<br>(0.28,<br>0.66)                                          | 0.56<br>(0.37,<br>0.71)                                                       | -                                                                   | -                                                                                |

|             |        |        |       |        |        |        |        |        |       |        |        |   |
|-------------|--------|--------|-------|--------|--------|--------|--------|--------|-------|--------|--------|---|
| log VL Day2 | 0.55   | 0.4    | 0.74  | 0.79   | 0.64   | 0.64   | 0.37   | 0.29   | 0.51  | 0.58   | 0.98   | - |
| PostChal-   | (0.35, | (0.17, | (0.6, | (0.68, | (0.47, | (0.47, | (0.14, | (0.05, | (0.3, | (0.39, | (0.96, |   |
| lenge       | 0.7)   | 0.59)  | 0.83) | 0.87)  | 0.77)  | 0.77)  | 0.57)  | 0.51)  | 0.67) | 0.73)  | 0.99)  |   |
| NasalSwab   |        |        |       |        |        |        |        |        |       |        |        |   |
| Subgenomic  |        |        |       |        |        |        |        |        |       |        |        |   |

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## Virus-Specific Antibodies by Viral Load

For numeric viral load, Spearman correlations with each virus-specific Ab will be calculated.

### 6.2.3m: (Male Only) Spearman Correlations of Virus-Specific Antibodies with Viral Load Measurements

|             | log<br>MN 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>MN<br>day<br>chal-<br>lenge | log<br>MSD<br>ECL 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>MSD<br>ECL<br>day<br>chal-<br>lenge | log<br>PsVNA<br>ID50 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>PsVNA<br>ID50<br>day<br>chal-<br>lenge | log<br>PsVNA<br>ID80 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>PsVNA<br>ID80<br>day<br>chal-<br>lenge |
|-------------|-------------------------------------------------|------------------------------------|---------------------------------------------------------|--------------------------------------------|------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|-----------------------------------------------|
| AUC BAL     | -0.53                                           | -0.67                              | -0.58                                                   | -0.61                                      | -0.54                                                      | -0.5                                          | -0.58                                                      | -0.6                                          |
| N1          | (-0.69,<br>-0.32)                               | (-0.79,<br>-0.51)                  | (-0.73,<br>-0.39)                                       | (-0.74,<br>-0.42)                          | (-0.7,<br>-0.34)                                           | (-0.66,<br>-0.28)                             | (-0.72,<br>-0.39)                                          | (-0.74,<br>-0.41)                             |
| AUC BAL     | -0.51                                           | -0.62                              | -0.55                                                   | -0.59                                      | -0.51                                                      | -0.44                                         | -0.56                                                      | -0.57                                         |
| Subgenomic  | (-0.68,<br>-0.31)                               | (-0.76,<br>-0.45)                  | (-0.7,<br>-0.35)                                        | (-0.73,<br>-0.4)                           | (-0.67,<br>-0.3)                                           | (-0.62,<br>-0.22)                             | (-0.71,<br>-0.36)                                          | (-0.72,<br>-0.38)                             |
| AUC Nasal   | -0.69                                           | -0.66                              | -0.73                                                   | -0.74                                      | -0.67                                                      | -0.62                                         | -0.68                                                      | -0.74                                         |
| Swab N1     | (-0.8,<br>-0.54)                                | (-0.78,<br>-0.49)                  | (-0.83,<br>-0.59)                                       | (-0.83,<br>-0.6)                           | (-0.79,<br>-0.51)                                          | (-0.75,<br>-0.44)                             | (-0.8,<br>-0.52)                                           | (-0.83,<br>-0.6)                              |
| AUC Nasal   | -0.72                                           | -0.73                              | -0.73                                                   | -0.74                                      | -0.73                                                      | -0.64                                         | -0.72                                                      | -0.76                                         |
| Swab        | (-0.82,<br>-0.57)                               | (-0.82,<br>-0.58)                  | (-0.83,<br>-0.59)                                       | (-0.83,<br>-0.6)                           | (-0.83,<br>-0.59)                                          | (-0.77,<br>-0.47)                             | (-0.82,<br>-0.57)                                          | (-0.85,<br>-0.63)                             |
| Subgenomic  |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| AUC OP      | -0.65                                           | -0.7                               | -0.72                                                   | -0.73                                      | -0.64                                                      | -0.58                                         | -0.68                                                      | -0.7                                          |
| Swab N1     | (-0.77,<br>-0.47)                               | (-0.8,<br>-0.54)                   | (-0.82,<br>-0.57)                                       | (-0.83,<br>-0.59)                          | (-0.77,<br>-0.47)                                          | (-0.73,<br>-0.39)                             | (-0.79,<br>-0.51)                                          | (-0.81,<br>-0.54)                             |
| AUC OP      | -0.56                                           | -0.61                              | -0.6                                                    | -0.64                                      | -0.57                                                      | -0.49                                         | -0.6                                                       | -0.6                                          |
| Swab        | (-0.71,<br>-0.36)                               | (-0.74,<br>-0.43)                  | (-0.74,<br>-0.41)                                       | (-0.77,<br>-0.47)                          | (-0.72,<br>-0.38)                                          | (-0.66,<br>-0.28)                             | (-0.74,<br>-0.42)                                          | (-0.74,<br>-0.41)                             |
| Subgenomic  |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| log VL Day2 | -0.41                                           | -0.53                              | -0.46                                                   | -0.49                                      | -0.43                                                      | -0.38                                         | -0.46                                                      | -0.49                                         |
| PostChal-   | (-0.59,<br>-0.18)                               | (-0.69,<br>-0.33)                  | (-0.63,<br>-0.24)                                       | (-0.66,<br>-0.28)                          | (-0.61,<br>-0.21)                                          | (-0.58,<br>-0.15)                             | (-0.63,<br>-0.24)                                          | (-0.66,<br>-0.27)                             |
| lenge BAL   |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| N1          |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| log VL Day2 | -0.45                                           | -0.54                              | -0.48                                                   | -0.51                                      | -0.46                                                      | -0.4                                          | -0.5                                                       | -0.51                                         |
| PostChal-   | (-0.63,<br>-0.23)                               | (-0.7,<br>-0.34)                   | (-0.65,<br>-0.26)                                       | (-0.67,<br>-0.3)                           | (-0.63,<br>-0.24)                                          | (-0.59,<br>-0.17)                             | (-0.66,<br>-0.29)                                          | (-0.67,<br>-0.3)                              |
| lenge BAL   |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| Subgenomic  |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| log VL Day2 | -0.39                                           | -0.4                               | -0.36                                                   | -0.39                                      | -0.41                                                      | -0.36                                         | -0.42                                                      | -0.41                                         |
| PostChal-   | (-0.58,<br>-0.16)                               | (-0.59,<br>-0.16)                  | (-0.56,<br>-0.13)                                       | (-0.58,<br>-0.15)                          | (-0.6,<br>-0.18)                                           | (-0.56,<br>-0.13)                             | (-0.6,<br>-0.19)                                           | (-0.6,<br>-0.18)                              |
| lenge       |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| OPswab N1   |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| log VL Day2 | -0.45                                           | -0.45                              | -0.43                                                   | -0.45                                      | -0.48                                                      | -0.43                                         | -0.49                                                      | -0.47                                         |
| PostChal-   | (-0.63,<br>-0.23)                               | (-0.63,<br>-0.23)                  | (-0.61,<br>-0.2)                                        | (-0.63,<br>-0.23)                          | (-0.65,<br>-0.26)                                          | (-0.61,<br>-0.21)                             | (-0.66,<br>-0.27)                                          | (-0.64,<br>-0.25)                             |
| lenge       |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| OPswab      |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| Subgenomic  |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| log VL Day2 | -0.78                                           | -0.77                              | -0.8                                                    | -0.77                                      | -0.82                                                      | -0.75                                         | -0.81                                                      | -0.82                                         |
| PostChal-   | (-0.86,<br>-0.67)                               | (-0.85,<br>-0.64)                  | (-0.87,<br>-0.68)                                       | (-0.86,<br>-0.65)                          | (-0.89,<br>-0.71)                                          | (-0.84,<br>-0.62)                             | (-0.88,<br>-0.7)                                           | (-0.89,<br>-0.72)                             |
| lenge       |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| NasalSwab   |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |
| N1          |                                                 |                                    |                                                         |                                            |                                                            |                                               |                                                            |                                               |

|             |         |         |         |         |         |         |         |         |
|-------------|---------|---------|---------|---------|---------|---------|---------|---------|
| log VL Day2 | -0.77   | -0.76   | -0.77   | -0.76   | -0.8    | -0.74   | -0.79   | -0.81   |
| PostChal-   | (-0.85, | (-0.85, | (-0.85, | (-0.84, | (-0.87, | (-0.83, | (-0.87, | (-0.88, |
| lenge       | -0.64)  | -0.63)  | -0.64)  | -0.62)  | -0.68)  | -0.6)   | -0.67)  | -0.7)   |
| NasalSwab   |         |         |         |         |         |         |         |         |
| Subgenomic  |         |         |         |         |         |         |         |         |

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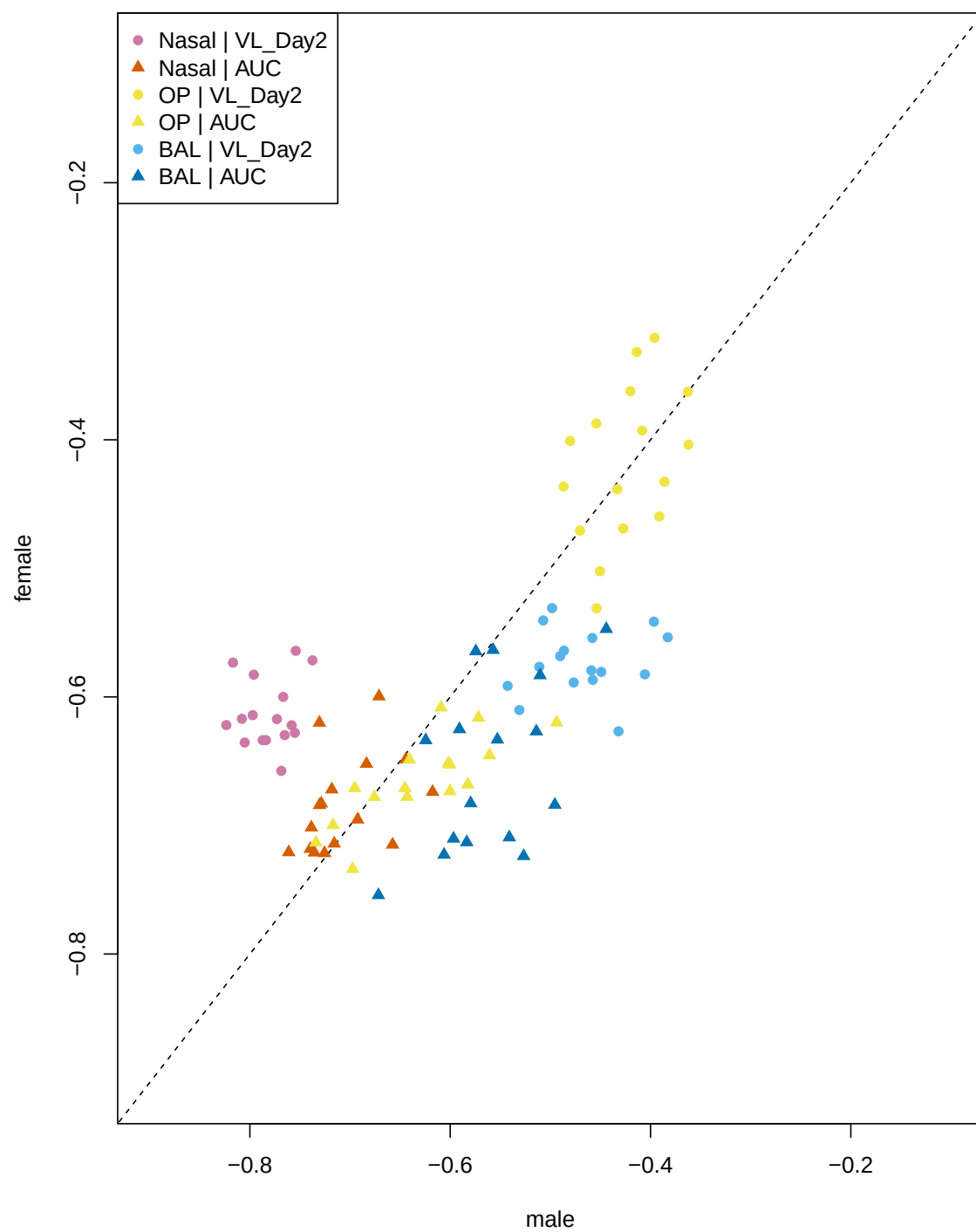
## 6.2.3f: (Female Only) Spearman Correlations of Virus-Specific Antibodies with Viral Load Measurements

|                        | log<br>MN 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>MN<br>day<br>chal-<br>lenge | log<br>MSD<br>ECL 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>MSD<br>ECL<br>day<br>chal-<br>lenge | log<br>PsVNA<br>ID50 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>PsVNA<br>ID50<br>day<br>chal-<br>lenge | log<br>PsVNA<br>ID80 2<br>weeks<br>prior<br>chal-<br>lenge | log<br>PsVNA<br>ID80<br>day<br>chal-<br>lenge |
|------------------------|-------------------------------------------------|------------------------------------|---------------------------------------------------------|--------------------------------------------|------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|-----------------------------------------------|
| AUC BAL                | -0.72                                           | -0.75                              | -0.71                                                   | -0.72                                      | -0.71                                                      | -0.68                                         | -0.68                                                      | -0.71                                         |
| N1                     | (-0.82,<br>-0.58)                               | (-0.84,<br>-0.62)                  | (-0.82,<br>-0.57)                                       | (-0.82,<br>-0.58)                          | (-0.81,<br>-0.56)                                          | (-0.8,<br>-0.53)                              | (-0.8,<br>-0.52)                                           | (-0.81,<br>-0.56)                             |
| AUC BAL                | -0.63                                           | -0.63                              | -0.63                                                   | -0.63                                      | -0.58                                                      | -0.55                                         | -0.56                                                      | -0.56                                         |
| Subgenomic             | (-0.76,<br>-0.45)                               | (-0.76,<br>-0.46)                  | (-0.76,<br>-0.46)                                       | (-0.76,<br>-0.45)                          | (-0.73,<br>-0.39)                                          | (-0.7,<br>-0.35)                              | (-0.71,<br>-0.37)                                          | (-0.71,<br>-0.37)                             |
| AUC Nasal              | -0.7                                            | -0.71                              | -0.68                                                   | -0.72                                      | -0.6                                                       | -0.67                                         | -0.65                                                      | -0.72                                         |
| Swab N1                | (-0.8,<br>-0.54)                                | (-0.82,<br>-0.57)                  | (-0.8,<br>-0.52)                                        | (-0.82,<br>-0.57)                          | (-0.74,<br>-0.41)                                          | (-0.79,<br>-0.51)                             | (-0.77,<br>-0.48)                                          | (-0.82,<br>-0.58)                             |
| AUC Nasal              | -0.71                                           | -0.72                              | -0.68                                                   | -0.7                                       | -0.62                                                      | -0.65                                         | -0.67                                                      | -0.72                                         |
| Swab                   | (-0.82,<br>-0.57)                               | (-0.82,<br>-0.58)                  | (-0.8,<br>-0.53)                                        | (-0.81,<br>-0.55)                          | (-0.75,<br>-0.44)                                          | (-0.77,<br>-0.48)                             | (-0.79,<br>-0.51)                                          | (-0.82,<br>-0.58)                             |
| Subgenomic             | -0.57                                           | -0.58                              | -0.53                                                   | -0.55                                      | -0.44                                                      | -0.48                                         | -0.51                                                      | -0.58                                         |
| AUC OP                 | -0.67                                           | -0.67                              | -0.7                                                    | -0.71                                      | -0.65                                                      | -0.67                                         | -0.68                                                      | -0.73                                         |
| Swab N1                | (-0.79,<br>-0.51)                               | (-0.79,<br>-0.51)                  | (-0.81,<br>-0.55)                                       | (-0.82,<br>-0.57)                          | (-0.77,<br>-0.48)                                          | (-0.79,<br>-0.5)                              | (-0.79,<br>-0.52)                                          | (-0.83,<br>-0.59)                             |
| AUC OP                 | -0.65                                           | -0.61                              | -0.65                                                   | -0.68                                      | -0.62                                                      | -0.62                                         | -0.65                                                      | -0.67                                         |
| Swab                   | (-0.77,<br>-0.47)                               | (-0.74,<br>-0.42)                  | (-0.78,<br>-0.48)                                       | (-0.79,<br>-0.52)                          | (-0.75,<br>-0.43)                                          | (-0.75,<br>-0.44)                             | (-0.77,<br>-0.48)                                          | (-0.79,<br>-0.51)                             |
| Subgenomic             | -0.47                                           | -0.42                              | -0.48                                                   | -0.52                                      | -0.43                                                      | -0.44                                         | -0.48                                                      | -0.51                                         |
| log VL Day2            | -0.58                                           | -0.61                              | -0.58                                                   | -0.57                                      | -0.63                                                      | -0.55                                         | -0.55                                                      | -0.56                                         |
| PostChal-<br>lenge BAL | (-0.73,<br>-0.39)                               | (-0.75,<br>-0.43)                  | (-0.72,<br>-0.39)                                       | (-0.72,<br>-0.37)                          | (-0.76,<br>-0.45)                                          | (-0.71,<br>-0.35)                             | (-0.71,<br>-0.35)                                          | (-0.71,<br>-0.37)                             |
| N1                     | -0.58                                           | -0.59                              | -0.59                                                   | -0.58                                      | -0.59                                                      | -0.54                                         | -0.53                                                      | -0.54                                         |
| log VL Day2            | (-0.73,<br>-0.39)                               | (-0.73,<br>-0.4)                   | (-0.73,<br>-0.4)                                        | (-0.72,<br>-0.38)                          | (-0.73,<br>-0.39)                                          | (-0.7,<br>-0.34)                              | (-0.69,<br>-0.32)                                          | (-0.7,<br>-0.34)                              |
| PostChal-<br>lenge BAL | -0.58                                           | -0.59                              | -0.59                                                   | -0.58                                      | -0.59                                                      | -0.54                                         | -0.53                                                      | -0.54                                         |
| Subgenomic             | (-0.73,<br>-0.39)                               | (-0.73,<br>-0.4)                   | (-0.73,<br>-0.4)                                        | (-0.72,<br>-0.38)                          | (-0.73,<br>-0.39)                                          | (-0.7,<br>-0.34)                              | (-0.69,<br>-0.32)                                          | (-0.7,<br>-0.34)                              |
| log VL Day2            | -0.46                                           | -0.32                              | -0.4                                                    | -0.43                                      | -0.33                                                      | -0.36                                         | -0.36                                                      | -0.39                                         |
| PostChal-<br>lenge     | (-0.64,<br>-0.24)                               | (-0.53,<br>-0.08)                  | (-0.59,<br>-0.17)                                       | (-0.61,<br>-0.21)                          | (-0.54,<br>-0.09)                                          | (-0.56,<br>-0.13)                             | (-0.56,<br>-0.13)                                          | (-0.58,<br>-0.16)                             |
| OPswab N1              | -0.46                                           | -0.32                              | -0.4                                                    | -0.43                                      | -0.33                                                      | -0.36                                         | -0.36                                                      | -0.39                                         |
| log VL Day2            | (-0.64,<br>-0.24)                               | (-0.53,<br>-0.08)                  | (-0.59,<br>-0.17)                                       | (-0.61,<br>-0.21)                          | (-0.54,<br>-0.09)                                          | (-0.56,<br>-0.13)                             | (-0.56,<br>-0.13)                                          | (-0.58,<br>-0.16)                             |
| PostChal-<br>lenge     | -0.53                                           | -0.39                              | -0.47                                                   | -0.5                                       | -0.4                                                       | -0.44                                         | -0.44                                                      | -0.47                                         |
| OPswab                 | (-0.69,<br>-0.33)                               | (-0.58,<br>-0.15)                  | (-0.64,<br>-0.25)                                       | (-0.67,<br>-0.29)                          | (-0.59,<br>-0.17)                                          | (-0.62,<br>-0.21)                             | (-0.62,<br>-0.21)                                          | (-0.64,<br>-0.25)                             |
| Subgenomic             | -0.53                                           | -0.39                              | -0.47                                                   | -0.5                                       | -0.4                                                       | -0.44                                         | -0.44                                                      | -0.47                                         |
| log VL Day2            | (-0.69,<br>-0.33)                               | (-0.58,<br>-0.15)                  | (-0.64,<br>-0.25)                                       | (-0.67,<br>-0.29)                          | (-0.59,<br>-0.17)                                          | (-0.62,<br>-0.21)                             | (-0.62,<br>-0.21)                                          | (-0.64,<br>-0.25)                             |
| PostChal-<br>lenge     | -0.63                                           | -0.6                               | -0.61                                                   | -0.62                                      | -0.57                                                      | -0.56                                         | -0.62                                                      | -0.62                                         |
| NasalSwab              | (-0.76,<br>-0.46)                               | (-0.74,<br>-0.41)                  | (-0.75,<br>-0.43)                                       | (-0.75,<br>-0.44)                          | (-0.72,<br>-0.38)                                          | (-0.71,<br>-0.37)                             | (-0.75,<br>-0.44)                                          | (-0.75,<br>-0.44)                             |
| N1                     | -0.46                                           | -0.41                              | -0.43                                                   | -0.44                                      | -0.38                                                      | -0.37                                         | -0.44                                                      | -0.44                                         |

|             |         |         |         |         |         |         |         |         |
|-------------|---------|---------|---------|---------|---------|---------|---------|---------|
| log VL Day2 | -0.66   | -0.62   | -0.63   | -0.63   | -0.58   | -0.57   | -0.63   | -0.64   |
| PostChal-   | (-0.78, | (-0.75, | (-0.76, | (-0.76, | (-0.73, | (-0.72, | (-0.76, | (-0.76, |
| lenge       | -0.49)  | -0.44)  | -0.45)  | -0.45)  | -0.39)  | -0.38)  | -0.46)  | -0.46)  |
| NasalSwab   |         |         |         |         |         |         |         |         |
| Subgenomic  |         |         |         |         |         |         |         |         |

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We plot the correlations from tables 6.2.3m and 6.2.3f to see if there is a pattern. It is important to realize that the correlation data are not independent points. To help visualize this, we plot the points by the type of viral load measurement. Specifically, we plot the points where color denotes location (the nasal viral loads are redish, the BAL are blue-ish, and the OP are yellow) and the shape is the time period of the measurement (circles are day 2 after challenge, triangles are the AUC [which is a weighted sum over several time points]). There are 16 points for each location/time-period set of points, representing the 8 virus-specific antibody measurements using either of the RT-qPCR assays (subgenomic or N1) for viral load measurement. Recall that within any specific location/time-period set, the two types of RT-qPCR assays (subgenomic or N1) are highly correlated (see Tables 6.2.2m and 6.2.2f).



In order to visualize the variability of that plot, we create 8 bootstrap data sets (sample each animal with replacement so that the total sample size is the same) and plot those bootstrap data sets in the same manner as the previous plot. Notice how the location/time-period set of points travel together and are variable (especially the OP|VL\_Day2 set).

